1) Rava - 1 cup

2) Sugar - 2 cups

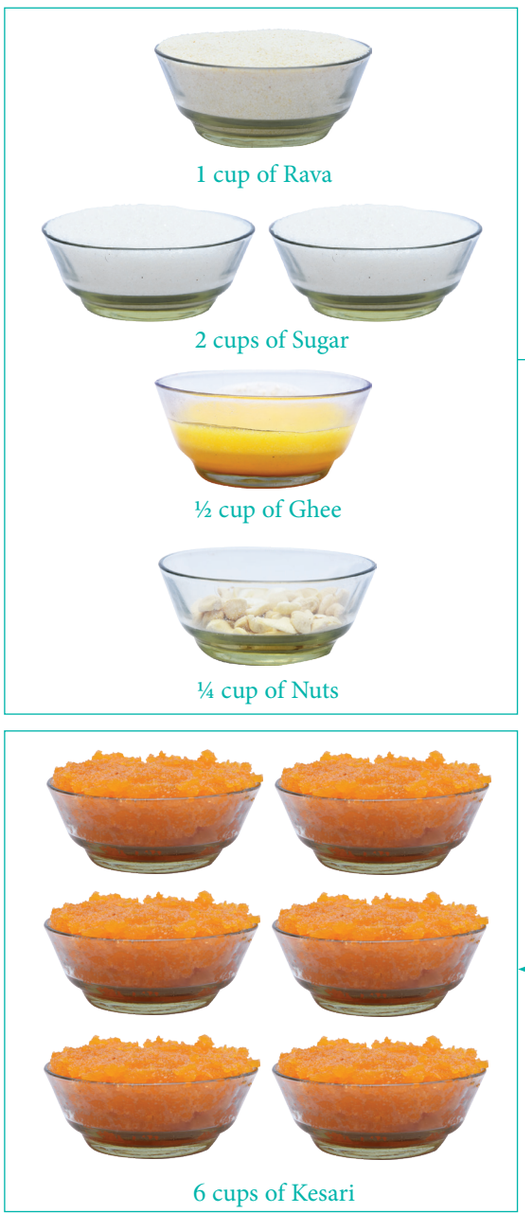
3) Ghee - $\frac{1}{2}$ cup

4) Nuts and Dry fruits - $\frac{1}{4}$ cup

Otherwise,

1 cup rava + 2 cups sugar + $\frac{1}{2}$ cup ghee + $\frac{1}{4}$ cup nuts and dry fruits → 6 cups kesari.

From the above information, we will be able to calculate the amount of ingredients that are required for the preparation of 3 cups of kesari as follows



$$\frac{1 \text{ cup rava}}{6 \text{ cups kesari}} \times 3 \text{ cups kesari} = \frac{1}{2} \text{ cup rava}$$

Alternatively, we can calculate the amount of kesari obtained from 3 cups rava as below.

$$\frac{6 \text{ cups kesari}}{1 \text{ cup rava}} \times 3 \text{ cups rava} = 18 \text{ cups kesari}$$

Similarly, we can calculate the required quantity of other ingredients too.

We can extend this concept to perform stoichiometric calculations for a chemical reaction. In Greek, *stoicheion* means element and *metron* means measure that is, **stoichiometry gives the numerical relationship between chemical quantities in a balanced chemical equation**. By applying the concept of stoichiometry, we can calculate the amount of reactants required to prepare a specific amount of a product and vice versa using balanced chemical equation.

Let us consider the following chemical reaction.

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

From this equation, we learnt that 1 mole of carbon reacts with 1 mole of oxygen molecule to form 1 mole of carbon dioxide.

$$1 \text{ mole of C} \equiv 1 \text{ mole of } O_2$$

$$\equiv 1 \text{ mole of } CO_2$$

The **symbol** '≡' means '**stoichiometrically equal to**'

11th Std Chemistry 001-037.indd 13 06-01-2021 15:02:50